

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458433.7712 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1444 +/- 0.0065
Transit depth (Rp/Rs)^2	2.09 +/- 0.19 [%]
Orbital Inclination (inc)	89.35 +/- 0.67
Airmass coefficient 1 (a1)	1.1619 +/- 0.0088
Airmass coefficient 2 (a2)	-0.138 +/- 0.012
Scatter in the residuals of the lightcurve fit is	0.76 %
Best Comparison Star	None
Optimal Aperture	4.64
Optimal Annulus	13.15
Transit Duration (day)	0.107 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1444 ± 0.0065 vs 0.131 (deviation 1.55σ)
Duration fit vs published	0.107 d vs 0.1075 d (deviation 0.18σ)
Mid-time O–C	-4.5 min
Depth SNR	16.8
Residual scatter	0.76 %

Light curve

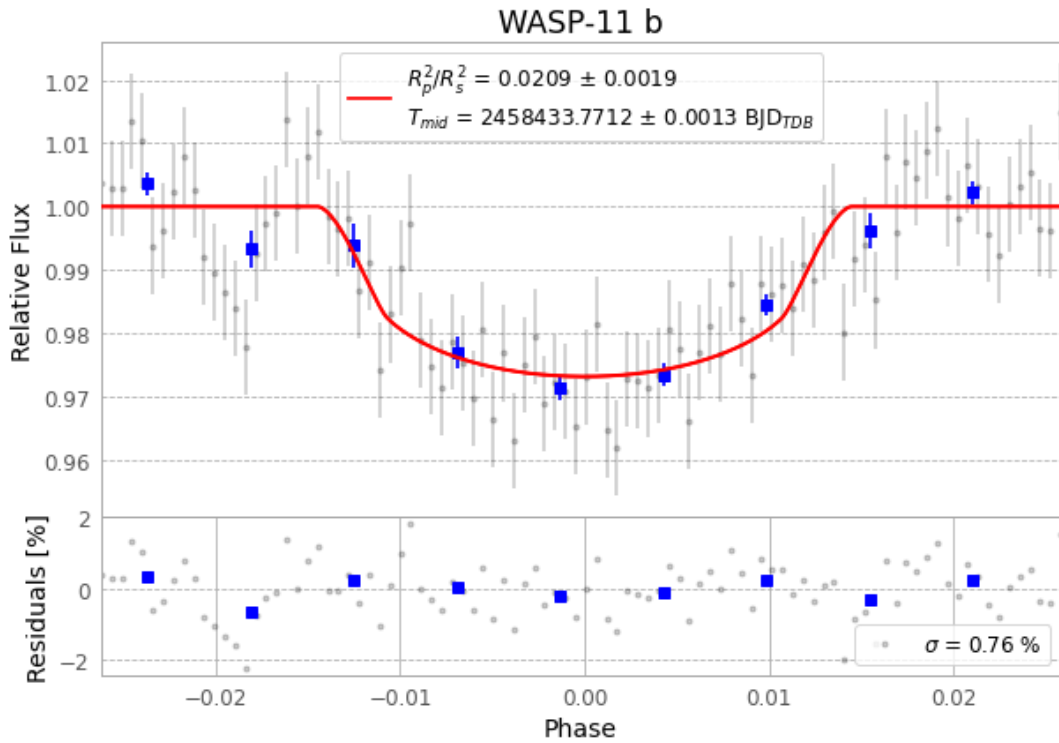


Figure 1 — FinalLightCurve_WASP-11 b_2018-11-10.png (SHA-256 e5166196e5241872...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 204], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 806e70cea371161e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

94 FITS frames (62 MB), HATP-10181111040012.FITS → HATP-10181111083912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-181111.log (c63ff52906f2...), FinalLightCurve_WASP-11 b_2018-11-10.png (e5166196e524...), FinalLightCurve_WASP-11 b_2018-11-10.csv (f7cedb69b756...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 00:15Z.